The University of Azad Jammu & Kashmir, Muzaffarabad

Department of Software Engineering

# Software Construction & Development (Lab)

## Lab 01

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| **Task Title:** | **Use Case Modeling & Scenario Analysis** |
| **Submitted to:** | **Engr. Dr. Asma Javaid** |
| **Student Name:** | **Feeza Kulsoom** |
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# Task 1: Online Ticket Reservation System

**Problem Statement:** Using the knowledge of online reservation systems, develop a Use Case Diagram that models the functional requirements for an Online Ticket Reservation System, including user interactions, booking workflows, and administrative controls.

## 1.1 System Actors & Identification

- **Passenger / Customer (Primary Actor):** Initiates the booking interaction, searches available travel routes, views seat layouts, books tickets, executes payment, and requests reservation cancellations.
- **System Admin (Secondary Actor):** Maintains system data, manages travel schedules, route timings, coach configurations, fare policies, and reviews cancellation requests.
- **Payment Gateway (External System Actor):** Third-party financial service responsible for validating credit/debit card information, authenticating transactions, and returning authorization tokens.

## 1.2 Core Use Cases & Relationships

- **Search Schedules & Availability:** Allows the passenger to query route origin, destination, and dates.
- **Select Seat & Class:** Permits seat selection across economy, business, or sleeper classes.
- **Book Ticket:** Core transaction use case creating the reservation instance.
- **Make Online Payment (<<include>>):** Mandatory sub-flow included during ticket booking to guarantee authorized payment clearing.
- **View / Download E-Ticket:** Generates printable ticket with QR code verification upon confirmed booking.
- **Cancel Reservation & Request Refund:** Allows passenger cancellation and initiates refund calculation.
- **Manage Routes, Fares & Schedules:** Administrative feature enabling CRUD operations on transit lines.

## 1.3 Use Case Diagram

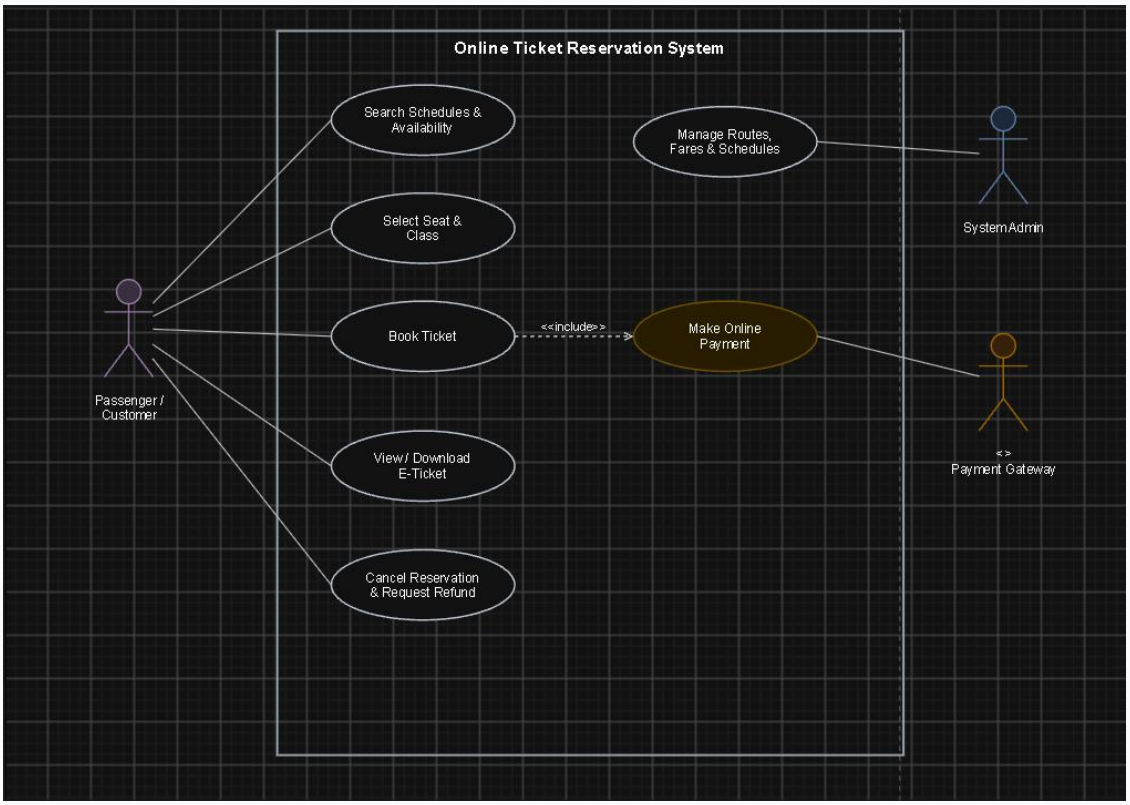



*Figure: Use Case Diagram - Online Ticket Reservation System*

# Task 2: Case Study – Credit Card Processing System

**Case Study Overview:** The Credit Card Processing System is used to purchase an item from any shop mall and maintain credit balance limits. The customer selects items, the vendor produces the bill, the card is swiped to authorize payment, and a balance statement/receipt is printed and signed.

## 2.1 Actor Identification

- **Customer (Primary Actor):** The purchaser who selects merchandise from the shopping mall, provides the credit card for transaction payment, and signs the merchant copy of the printed receipt.
- **Vendor / Cashier (Secondary Actor):** The merchant representative who totals selected items, generates the sales bill, initiates the card swipe on the POS terminal, and issues the receipt.
- **Credit Card Machine / Bank Server (System Actor):** The electronic terminal and banking clearinghouse that communicates card data, verifies balance limits, validates credit availability, and prints the transaction statement.

## 2.2 Use Case Identification & Descriptions

- **Select Item(s):** Customer browses store aisles and selects desired items for checkout.
- **Generate Bill:** Vendor computes unit prices, applies taxes/discounts, and calculates the total payable amount.
- **Swipe / Insert Card:** Vendor/Customer interfaces card chip or magnetic stripe with the terminal reader.
- **Process Transaction & Check Balance Limit (<<include>>):** Machine verifies credit limitation against the issuing bank and reserves transaction funds.
- **Print Balance Statement / Receipt (<<include>>):** Machine produces paper vouchers with transaction status and updated credit details.
- **Sign Receipt:** Customer affixes physical signature to the vendor receipt copy for authorization verification.

## 2.3 Use Case Diagram

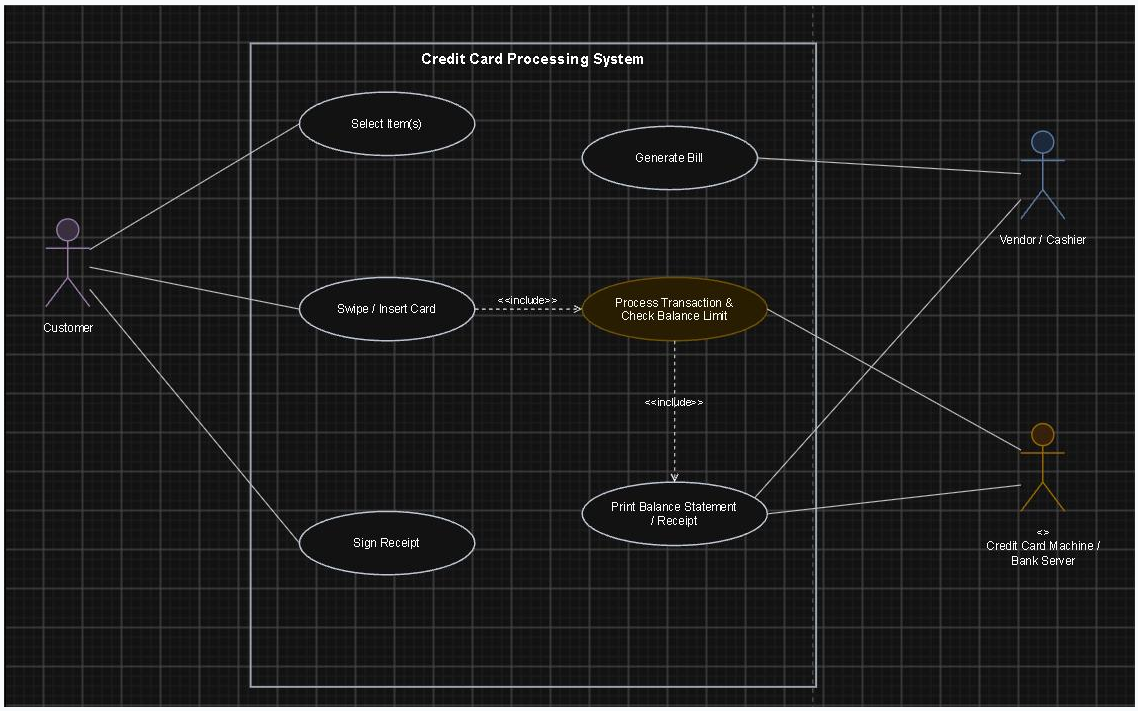



*Figure: Use Case Diagram - Credit Card Processing System*